

Design Proposal: Group Index Extraction and Advanced Filtering using Ring-Assisted Mach-Zehnder Interferometers

edX Username: nnestorhs

Phot1x: Silicon Photonics Design, Fabrication and Data Analysis

Abstract—This design proposal presents the development of silicon photonic interferometric circuits aimed at extracting the waveguide group index (n_g) and exploring advanced spectral filtering. The core design is a Ring-Assisted Mach-Zehnder Interferometer (RAMZI) incorporating a microring resonator coupled to each arm of the MZI. Four variations of the RAMZI are proposed to investigate the impact of ring coupling coefficients and phase conditions. A standard MZI with an equivalent base delay is included as a reference structure to reliably extract the group index. The report details the theoretical framework, mathematical models, and simulated performance of these devices, which will be fabricated to compare theoretical predictions with experimental data.

I. INTRO

Precise characterization of waveguide dispersion and the realization of compact, high-performance filters are critical in integrated silicon photonics. The primary design objective of this project is twofold: (1) to extract the waveguide group index using a standard unbalanced interferometer, and (2) to design and analyze a custom Ring-Assisted Mach-Zehnder Interferometer (RAMZI) architecture for advanced spectral shaping.

The proposal utilizes standard strip waveguides (350 nm width, 220 nm height) operating in the quasi-TE polarization. The standard MZI serves as the baseline for parameter extraction, while the custom RAMZI—incorporating a microring resonator in both the upper and lower arms of the MZI—explores the transition from standard sinusoidal interference to complex, tailorable resonant responses (e.g., flat-top filters).

II. THEORY

A. Standard MZI and Group Index Extraction

For the reference unbalanced MZI with a path length difference $\Delta L = L_2 - L_1$, the Free Spectral Range (FSR) is directly related to the group index (n_g):

$$FSR_{MZI} \approx \frac{\lambda^2}{n_g \Delta L} \quad (1)$$

From this relationship, the group index will be experimentally extracted using the measured FSR:

$$n_g = \frac{\lambda^2}{FSR_{MZI} \cdot \Delta L} \quad (2)$$

B. Ring-Assisted MZI (RAMZI)

The custom RAMZI design introduces a microring resonator (radius R) coupled to each arm (L_1 and L_2) of the MZI. The transmission of a single all-pass microring resonator is given by:

$$T_{ring} = \frac{a - \rho e^{-j\phi_r}}{1 - a\rho e^{-j\phi_r}} \quad (3)$$

where a is the round-trip transmission (loss), ρ is the self-coupling coefficient (related to the power coupling κ^2 by $\rho^2 + \kappa^2 = 1$), and $\phi_r = 2\pi n_{eff}(2\pi R)/\lambda$ is the round-trip phase.

With a ring in each arm, the overall transfer function of the RAMZI depends on the interference of the two dispersive paths. The output intensity can be modeled as:

$$I_{out}(\lambda) = \frac{I_{in}}{4} |e^{-j\beta L_1} T_{ring,1} + e^{-j\beta L_2} T_{ring,2}|^2 \quad (4)$$

where $\beta = 2\pi n_{eff}/\lambda$, and $T_{ring,1}$ and $T_{ring,2}$ are the transfer functions of the respective rings. By symmetrically or asymmetrically tailoring the properties of each ring (e.g., introducing a slight radius detuning or varying the coupling coefficient κ), the interference pattern can be sculpted to create broadened flat-top responses.

III. MODELLING AND SIMULATION

A. Waveguide Geometry and Mode Profile

The designs are based on standard silicon strip waveguides:

- **Width:** 350 nm
- **Height:** 220 nm
- **Polarization:** Quasi-TE

B. Effective and Group Index

Simulations were performed using Lumerical MODE to extract the dispersion characteristics across the O-band.

The compact model describing the effective index is:

$$n_{eff}(\lambda) = n_0 + n_1(\lambda - 1.31) + n_2(\lambda - 1.31)^2 \quad (5)$$

TABLE I
POLYNOMIAL FITTING PARAMETERS FOR EFFECTIVE INDEX.

Parameter	Values
n_0	2.41624
n_1	$-1.40182\mu m^{-1}$
n_2	$-0.311122\mu m^{-2}$

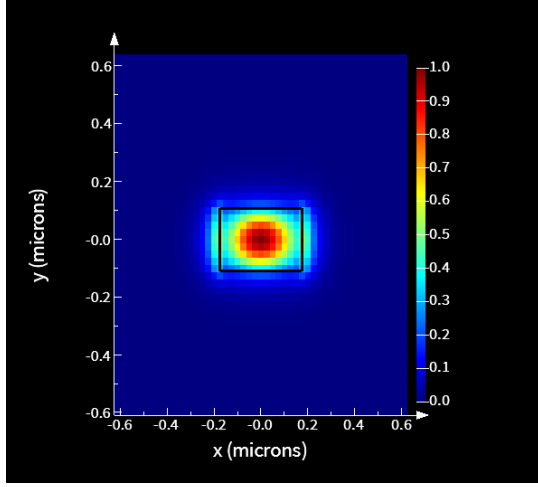


Fig. 1. Simulated quasi-TE fundamental mode profile from Lumerical MODE Solutions.

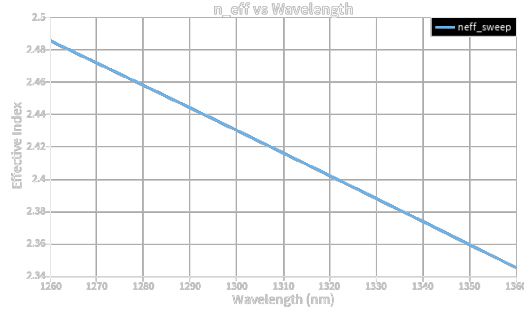


Fig. 2. Plot of simulated effective index (n_{eff}) and group index (n_g) versus wavelength.

C. Parameter Variations

A total of five designs are proposed: one reference MZI and four RAMZI variations. The base delay (ΔL) for the MZI and the RAMZI backbone is kept constant to ensure the base FSR fits within the 50 nm measurement window. The variations explore different power coupling coefficients (κ^2) between the bus waveguides and the microrings to observe under-coupled, critically coupled, and over-coupled regimes. For each ring, the total length must be adjusted properly to account for double the FSR from the MZI and the addition of a fraction of a pi shift to align the ring filter response to the MZI filter response.

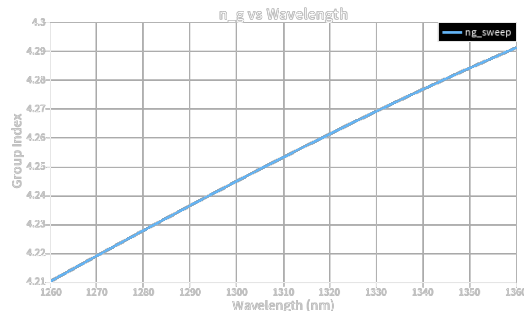


Fig. 3. Plot of simulated group index (n_g) versus wavelength.

TABLE II
CIRCUIT PARAMETER VARIATIONS

Design	ΔL (μm)	Ring coupling gap (nm)	Coupling length (μm)
1	20.538	N/A	N/A
2	20.538	$g_1 = 80, g_2 = 160$	$L_{c1} = 4, L_{c2} = 4$
3	20.538	$g_1 = 100, g_2 = 140$	$L_{c1} = 4, L_{c2} = 4$
4	41.075	N/A	N/A
5	41.075	$g_1 = 80, g_2 = 160$	$L_{c1} = 5, L_{c2} = 5$
6	41.075	$g_1 = 100, g_2 = 140$	$L_{c1} = 5, L_{c2} = 5$

D. Circuit Transmission Spectrum

The full circuits were simulated using Lumerical INTERCONNECT to verify the transfer functions. The dual-ring RAMZI designs demonstrate how the interaction between the MZI delay and the coupled resonators reshapes the spectral response.

IV. FABRICATION

[To be completed later to include layout and details about the fabrication process.]

V. EXPERIMENT DATA

[To be completed later once experimental results are received.]

VI. ANALYSIS

[To be completed later. This section will compare the experimentally extracted n_g from Design 1 with simulated values, and analyze the coupling and detuning regimes of the fabricated RAMZI circuits (Designs 2-5).]

VII. CONCLUSION

[To be completed later summarizing the findings of the fabrication and testing.]

REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge: Cambridge University Press, 2015.